SCORING FOR GRID CONTROLLERS – PRELIMINARY CONSIDERATIONS OF ACTION, GESTURE, FUNCTIONALITY, AND LIMITATIONS

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ABSTRACT

In this paper and performance, I aim to demonstrate a set of considerations and influences on a type of score designed for and specific to the 8x8 grid controller commonly associated with software such as Ableton Live.

The grid controller offers a unique method of control that is unlike other MIDI controller systems, and the problem of scoring for it comes primarily from the flexibility of functionality that the controller offers: from mapping to a grid layout of 'clips' within Ableton Live, to conventional MIDI-note systems. The deficiency with this latter mode is the requirement to learn a functionally non-linear map of pitch that is somewhat unintuitive in relation to the traditional staff-based hierarchy.

Through the performance and discussion of the score itself, this paper will show some of the idiosyncratic characteristics of an 8x8 grid controller, possible scoring paradigms, limitations shown thus far and potential limitations moving forward, and further considerations for new methods of working with this grid system.

1. INTRODUCTION

This paper and its accompanying performance outlines the influences and considerations in developing the composition Dead Dirt (Giles 2015) for (augmented) flute and computer performer, utilising an idiosyncratic notational system based on a spatio-temporal design for an 8x8 Novation Launchpad grid controller with Ableton Live.

There is a rich history throughout the 20th and 21st centuries of very specific notation conventions being developed for specific works, or specific players, or specific instruments; a type of notation that is best described as idiosyncratic – a mode peculiar to an individual. Composers have approached the notation for acoustic instruments and electronics in ways that are increasingly individual not only to the composer and the work, but the player for whom it is written and indeed, the instrument itself. The following section shows some examples of instrument-specific notation, and notation for electronics, looking for methods that may be relatable to a method of scoring for a grid controller.

1.1. Overview of Approaches to Notational Movement/Gesture-Mapping

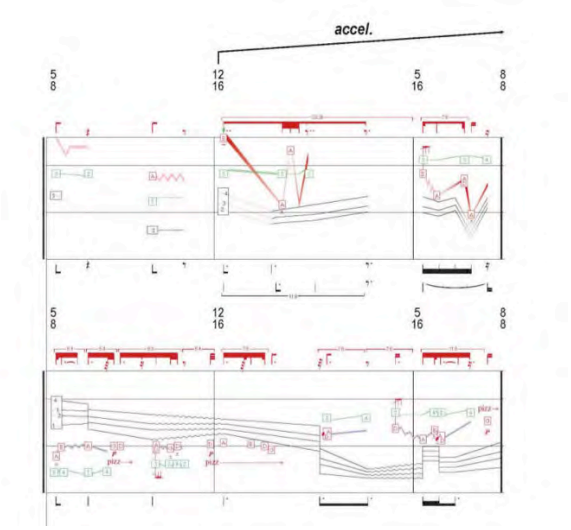


Figure 1. Aaron Cassidy, *Second String Quartet* (2009/10). Mm. 57-59. Violins 1 & 2.

Figure 1 shows a short section of Aaron Cassidy's *Second String Quartet*, in which the composer has utilised a physical mapping of the instruments in question in a type of tablature notation that shows the physical movement necessary in the piece – the performative gesture – rather than the resultant sonic output, which is variable. Cassidy (2013) explains that the staff lines represent, from top to bottom, the bridge of the violin, the end of the fingerboard, the end of the body, and the nut, with bow direction and intensity indicated in green, bow movement across the strings in red, and finger movement across the strings in black. These actions are all rhythmically independent of one another, and the resultant sound is unstable and indeterminate due to variations in bow quality and particularly hand size.

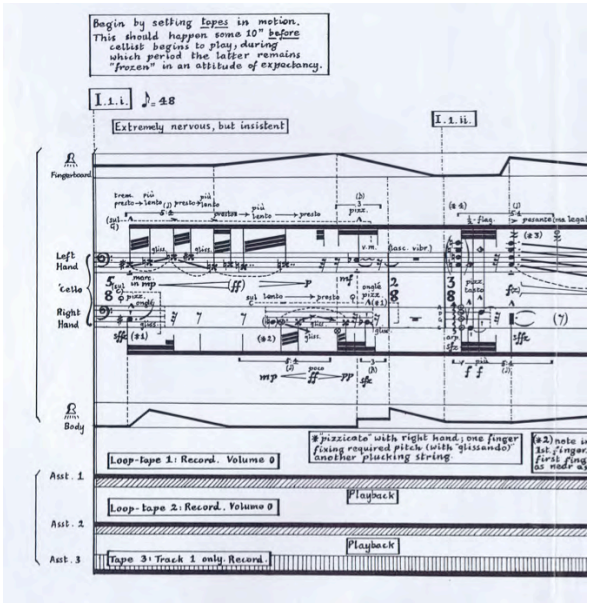


Figure 2. Brian Ferneyhough, *Time and Motion Study II* (1978). Mm. 1-3.

Ferneyhough's *Time and Motion Study II* for solo cellist with amplified cello and eight-speaker sound system (**Figure 2**) illustrates an earlier (than Cassidy) approach to the mapping of physical movement and its influence on sound production. It also shows a type of notation describing the electronic elements of the music, their durations, and the increase and decrease of amplitude over time. Breaking the score down into systems from low to high is the first three tape/computer loops being switched on and recording, with an instruction to enable playback. Notated above and below the cello-proper are the foot-controller envelopes for a microphone placed under the cello's fingerboard, and on the body of it, which the cellist controls (see: (Still 2014) for more detailed information on the technical setup). The cellist needs to control those pedals, play extraordinarily intricate parts on the cello itself, and at some points sing (as outlined in (Iddon 2006)); this bodily interaction with and against the cello is mapped very clearly through the notation and, indeed, the interaction of the body with the instrument is necessary for the piece to be effective as there is a palpable sense of tension in the performance of the work through the cellist's physical actions. Unlike the Cassidy piece in **Figure 1**, this score outlines the *resultant* sound, but the mechanisms for creating that sound are still very much contained within the notation through the decoupling of the right and left hand, as shown in **Figure 2**.

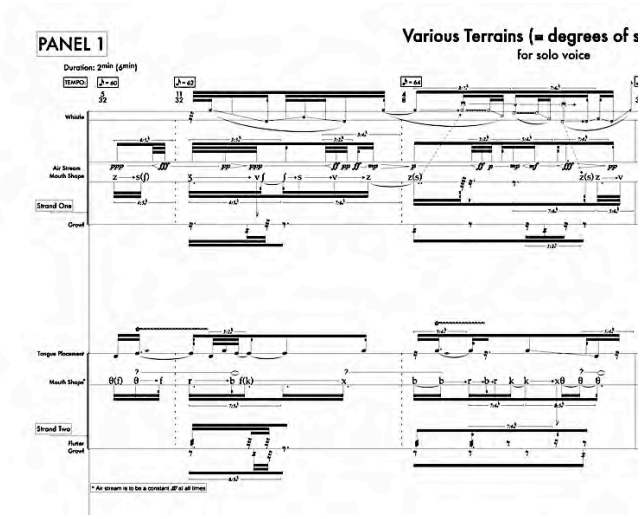


Figure 3. Michael Baldwin, *Various Terains (= degrees of similarity)* (2011). Mm. 1-3, threads 1 & 2.

Various Terrains shows a somewhat similar approach to both of the above examples, with the rhythmic independence of different sound production of a solo voice notated across the independent staves in each system. In practice there are a few possible realisations of the piece, based on the order in which the strands are performed. **Figure 3** only contains the first few bars of the first two strands of the first panel of the score, however it serves to illustrate the importance of mapping the action itself rather than the result of the action not only for clarity, but also for novel and exciting output. The top system (strand 1) contains a whistled sound, the singer's air pressure (and thus dynamic) along with the type of sound to be produced, and finally a growl. It is the interaction and instability of all of these actions that produce the desired effect in performance. It is an interesting example because it combines the extreme decoupling of actions akin to the Cassidy, and hence the instability, with the extreme determinism of the result of that action akin to the Ferneyhough.

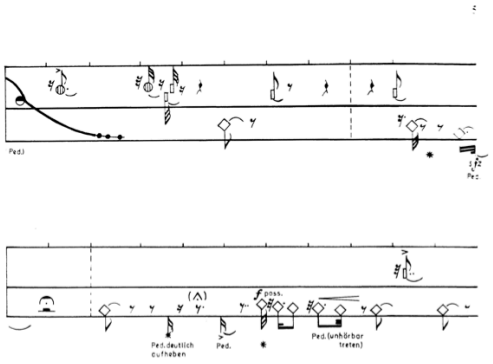


Figure 4. Helmut Lachenmann, *Guero* (1988). Mm. 1-3.

Lachenmann's *Guero*, for solo piano, takes a novel approach to the topography of the instrument, mapping different types of gesture across the surface of the keyboard without depressing the keys themselves, resulting in a percussive sound akin to the folk instrument of the same name. **Figure 4**, this notation is a

map of the movement required to produce the desired sound, but unlike the Cassidy the movement is completely deterministic – there are no variables such as hand size – and the resultant sound will be within expected margins of difference from performance to performance. Unlike the Cassidy, the movement does *not* produce an indeterminate sound as a result of the activity. *Guero* is included in this paper as an example that is very close to the desired results of *Dead Dirt* as a scoring paradigm, which will be discussed further below.

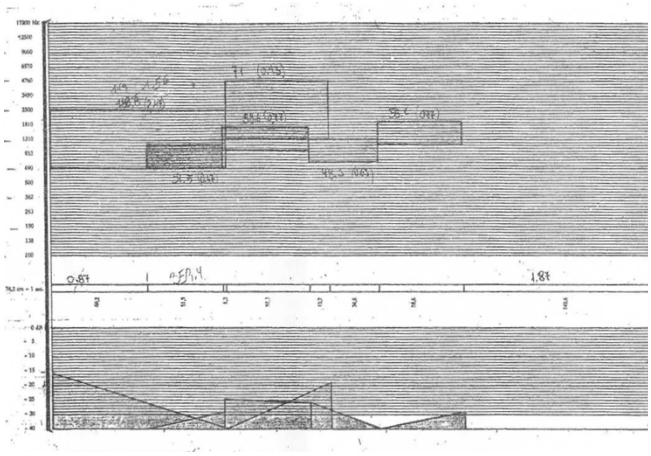


Figure 5. Karlheinz Stockhausen, *Nr. 3 Electronische Studien, Studie II* (1956). p. 1.

The final example of interesting and individualistic approaches to scoring is Stockhausen's *Elektronische Studie II*, which has famously been realised and distributed with MaxMSP for a number of years as a demonstration patch. The excerpt in **Figure 5** is from the original score rather than the Max realisation, and shows Stockhausen's approach to mapping blocks of frequencies (the top part of the system) and their amplitude envelopes (the bottom part of the system) over time (horizontally). While not performative, this is an interesting example of formalism in electronic music, and the gestures created in realisation have an incredible sense of physicality to them, akin to the feeling created by a performance of the Ferneyhough piece in **Figure 2**[1].

As illustrated through these excerpts, in their own right a very small snapshot of the varied approaches to composing for specific instruments and their idiosyncrasies, any approach to notating for a grid controller should be similarly idiosyncratic. Because grid controllers have a non-linear and variable functionality for each of the 64 pads, they provide the ability to reach great distances across octaves without shifting the register of the pads, or without physically reaching long distances. However, the linear visual association with pitch that is common with the piano may also not be the most useful or efficient form of notation for this controller, especially if the computer performer is re-training on this new instrument/controller from another instrument. Not to mention the relevance and directness of expressing something triggered with staffed pitch and whether that has an impact on the inherent semantics of the score. Of course, much of this is dependent upon the outcomes that are desired, and consideration must be made as to whether the grid controller is the best device for the composition at hand.

2. IDIOSYNCRATIC FEATURES OF GRID CONTROLLERS

In light of the above, it seems relevant to work towards unpacking the obvious physical features that distinguish a grid controller from a piano controller, or a set of faders, or an XY touchpad. By making these observations of the physicality of the controller, a composer may start to think about specific notational conventions for that design, in much the same way that Cassidy exploits the physical design of the violin in **Figure 1** and how Lachenmann utilises the physical characteristics of the piano in his composition *Guero* (**Figure 4**) and these instrument characteristics enter into the semantic qualities of the score itself – thus, an idiosyncratic notation for the individual characteristics of the instrument for which the piece is being composed.

2.1. Topographic Layout

The most evident design characteristic of the grid controller is its topographic layout, which relates to the next subheading (N-Dimensional layout), this layout allows for the mapping of triggers (in the form of pitch or any other MIDI message) within a tightly confined physical space. The design of these controllers, while novel in their application, draws upon a history of grid-designs for music dating back to the 1980s in software, and much earlier in human interface design for things like games and even cities (Adeney 2011); all of which demonstrate a degree of non-linear spatial interactivity.

Potentially, every 12 pads from bottom left to top right could represent each of the pitches of the chromatic scale, enabling rapid jumps from one register to another across multiple octaves with only one hand. But more powerfully, it allows the subdivision of the grid into four sets of sixteen pads, enabling simultaneous independent control of different sounds within a DAW – drums, piano, harmonies, and so on, enabling immediate single-performer control of a range of triggered parameters. The flexibility of triggered events within one confined geographic space is immensely powerful, and when combined with the ability to 'shift' the 'register' using the touch of buttons at the top of the controller, this potentially enables very rapid changes between preset groupings of triggers.

[1] This is the opinion of the author at the time of writing, of course other people may not agree.



Figure 6. Novation Launchpad; http://global.novationmusic.com/sites/default/files/styles/cta_scale_640/public/7b-LP-Pan_0.png

There are other instruments that have a topographic layout, all keyboard instruments for example, and even guitars, lutes and similar could be seen as having a topographic layout. None except perhaps lutes offer such a range of pitch in one physical hand space, and none at all offer the range of flexibility for *use* of the space that an 8x8 grid offers. For comparison, a clarinet or a trumpet does not have a topographic layout due to the variability of key or valve depression to achieve linearity of pitch structures.

2.2. N-Dimensional Layout

Strongly linked to the idea of the topographic layout described above is the idea of an n-dimensional layout, although in reality this is limited to only two dimensions (in the Novation Launchpad and Launchpad-S, and three dimensions in the Launchpad-Pro and Ableton Push controller[1]), the X – or horizontal axis – and the Y – or vertical axis. From a conceptual perspective this allows for conceptions of layout in any combination of right to left, left to right, top to bottom or bottom to top, all of which could be interlaced and varied depending on the requirement of the piece and in consideration of the two-dimensional limitation, ignoring for the moment the third dimension of velocity in more sophisticated controllers. This notion of thinking about n-dimensional space has a resonance with Earle Brown's conception of music as activity within a space, and also with the mappings of gesture and movement (both discussed below) evident in the excerpts in the previous section.

Brown (1961) states in the score for *Four Systems*:

> … to have elements exist in space … space as an infinitude of directions from an infinitude of points in space … to work (compositionally and in performance) to right, left, back, forward, up, down, and all points between … the score [being] a picture of this space at one instant, which must always be considered as unreal and/or transitory … a performer must set this all in motion (time), which is to say, realize that it is in motion and step into it … either sit and let it move or move through it at all speeds.

With the additional note: "[coefficient of] intensity and duration [is] space forward and back."

These quotes are accompanied by an image of three-dimensional space represented by a cube with branching dimensions within it, and Brown mentions the instigation of time, both literally and metaphorically, as a fourth dimension within this space. Metaphorically due to the implications of n-dimensional space in music, but literally in that four dimensions consist of the established three: X, Y, and Z, plus *t* for time, that we inhabit as humans.

This n-dimensional space relates strongly to the grid controller, because you could conceive of the 'start' of the device as any point on the X–or–Y-axes and their junctures, which is *not* a characteristic of other controller types, and, linked with the topographic layout described above, is one of the characteristics available for consideration in a scoring system for grid controllers. In the section that follows, I will describe the approach taken in the composition *Dead Dirt* (Giles 2015), the considerations of gesture mapping and movement mapping, the limitations of the controller and the composition itself in relation to a grid controller, and how the inherent characteristic of triggering and pre-designed sound plays a strong role in the compositional layout. I will also describe how considerations of n-dimensional space influenced the layout of the score, particularly as it relates to the time domain.

3. DEAD DIRT: EXAMINATION OF NOTATION SYSTEM FOR GRID CONTROLLER

3.1. Background to the Piece

Dead Dirt was composed for an exhibition opening at the Carlton Connect Initiative in Melbourne, Australia, of the same name. The theme of the exhibition was the notion that our connection with soil is diminishing, and public awareness of the importance of soil on agriculture and life is minimal.

My composition responds to the same thing, utilising data of soil analysis including levels of oxygen, nitrogen, and other compounds as connection points between the nodes in the electronic part as shown in **Figure 7**; it is not a literal transcription of data into music or sound, but an adaptation of the data for artistic purposes, leading to structures that would otherwise not have arisen. I won't detail the process of creation; rather

[1] These dimensions refer to 'musical' dimensions, or the crossover of 'musical' dimensions such as pitch and time, with the physical dimensions of X and Y. Specifically, the Launchpad Pro and Ableton Push both have velocity sensitivity on the pads, combining the two physical dimensions of X and Y with the third 'musical' dimension of amplitude.

I will discuss the mapping of movement in the structure of *Dead Dirt*.

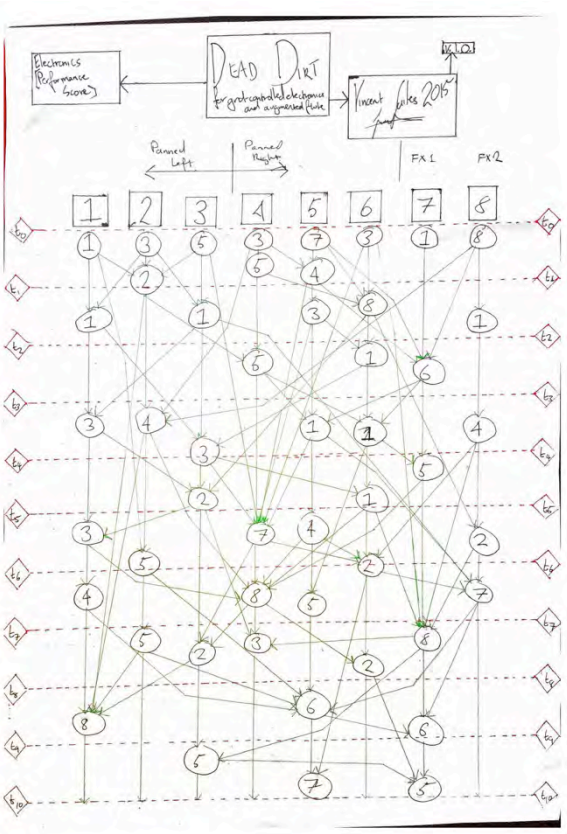


Figure 7. *Dead Dirt*, electronic part (Giles, 2015)

3.2. Mapping Movement

In the construction of the electronic part of *Dead Dirt,* the performance score specifies indeterminate time durations indicated as t_0, t_1, t_2,…,t_9, t_{10} on the left and right of vertical axis. The first row: [1], [2], [3], … , [8] represents each of the eight columns of pads on the 8x8 grid controller, with a network of 'nodes' moving down the page chronologically. Each node contains a number which represents the row correlating to the column it aligns with; for example, the 3 in the first column says that a little while after t_3, the 3rd row in the first column of pads is to be activated. Whatever event is associated with that pad then becomes active until the movement to the next node.

On most nodes there is a choice of three movements to the next node, which is to be triggered at the relevant point in time which the player should simultaneously deactivate the current node. For example, moving from the same 3 mentioned above to the 8 in column 4 requires the 3 to be deactivated upon activation of the 8. The instruction from the score states: "[o]nce a node is moved on from, the preceding node must be silenced." (Giles 2015)

This 'node'-style score layout can loosely be seen as having been influenced by the Centre for Electronic Media Art music software *Nodal*, an outline of which can be found in (McCormack et al. 2005), and more relevantly, the spatio-temporal concepts that precede Nodal discussed in (McIlwain and Pietsch 1996). This relationship between space and time is evident in my approach the score not only in the electronics, but the flute part too (see **Figure 8**).

Mapping movement across the controller in this way enables a sense of interactivity not only with the controller but also with whatever the pre-determined material happens to be. In treating the triggering of pre-made material as a physical movement; a map of the physical space the controller inhabits that can be negotiated by the performer. This approach echoes those taken by Cassidy and particularly Lachenmann in the excerpts above (**Figure 1** and **Figure 4**) approaching the idiosyncrasies of the instrument (or controller) and trying to approach it for what it is and what it can do, rather than forcing it to a pre-existing idea of notation and the associated hierarchies that come with that, thus mapping a spatial configuration (the X and Y) over time.

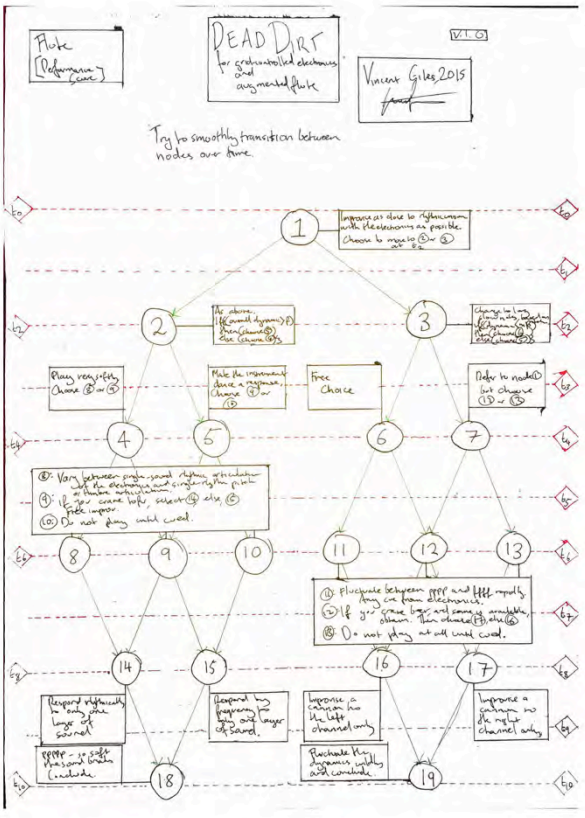


Figure 8. *Dead Dirt*, flute part (Giles 2015)

Figure 9 describes each of the hand-written boxes of instruction in **Figure 8**:

1: Improvise as close to rhythmic unison with the electronics as possible. Choose to move to **2** or **3**.

2: As above.
```
If { (overall dynamic > f)
Then (choose 5)
Else (choose 4)}
```

3: Change to long, slow notes based on frequency/pitch.
```
If { (overall dynamic < mp)
Then (choose 6)
Else (choose 7) }
```

4: Play very softly. Choose **8** or **9**.

5: Make the instrument *dance* a response. Choose **9** or **10**.

6: Free choice.

7: Refer to node **1**, but choose **12** or **13**.

8: Vary between single sound rhythmic articulation of the electronics, and single-rhythm pitch articulation.

9: Free improvisation: if you crave tofu, select **14**, else, **15**.

10: Do not play at all, until cued.

11: Fluctuate between *pppp* and *ffff* rapidly in response to any cue from the electronics.

12: If you crave beer, and some is available, go and obtain, then choose **17**, else choose **16**.

13: Do not play at all, until cued.

14: Respond rhythmically to only one layer of sound.

15: Respond by frequency/timbre to only one layer of sound.

16: Improvise a cannon with the left channel only.

17: Improvise a cannon to the right channel only.

18: *ppppp* – so soft that the sound breaks. Conclude the piece.

19: Fluctuate the dynamics wildly, and conclude the piece.

Figure 9. Clarified node-instructions for flutist in *Dead Dirt*.

3.3. Limitations of Dead Dirt

Dead Dirt shows a number of limitations, the most prominent of which is the limited interactivity *directly with the material at the time of production*, and the second is the requirement for pre-made sound to be linked to the trigger point. In the case of this piece being a semi-improvised work that is functionally for (augmented) flute and electronics, the need for the electronics to be fully interactive for the computer performer was not there. However, in considering the potential of the grid layout as a performative system, then certainly some consideration on how to overcome this limitation is worthwhile.

3.3.1. Triggering, Interactivity, and Pre-composition

The lack of direct interaction with the produced material in favour of simply triggering pre-made material, as mentioned above, is a function of this particular piece. However it's entirely conceivable that a similar movement map could be employed in a directly interactive composition without regard for the MIDI note or any other controller data: in this way taking the notion of mapping movement in a 2-dimensional space as a generator of something directly tangible. Whether or not this is particularly effective technique for the type of controller will need to be explored further, however, the inclusion of the velocity dimension in newer models of the controller may help. In the case of *Dead Dirt* however, the triggering system that is employed is highly effective as the material that is pre-designed and organised within Live's 'session view' (**Figure 10**) is itself highly gestural and interactive. The navigation of the 3-dimensional space-time notation then becomes semi-deterministic (or choose-your-own-adventure) movement through this pre-designed material, giving rise to emergent structures against which the flutist must improvise according to her own set of specific instructions.

The notion of interactivity therefore comes not from an interaction between the computer performer and their own material, but between the flutist and the electronic material; or between human and human-controlled-machine. In a sense, the computer performer therefore becomes a semi-independent agent negotiating a predetermined structure with variables, while the flutist is granted far greater agency and responds not to the computer performer, but to the triggered material – somewhat like a more traditional piece for acoustic instrument and tape/playback. With this design, the pre-designed material needs to be strong and therefore provide something for the flutist to interact with.



Figure 10. Screenshot of the basic 8x8 grid layout used in *Dead Dirt*, without content in columns 7 and 8[1].

With all of this in mind, a brief mention of the importance and considerations when creating the electronic sounds seems pertinent. First, the source for the electronics is sonification of soil-related data, though a different dataset to that used in the development of the score. Describing the full process of sonification falls outside this paper, but once sonified, the wave files form the basis of all the electronic material through manipulations, transcriptions into MIDI and novel sound packs contained within Live, and again transformed into audio for further manipulation. From there, a spatial element is added (as seen at the bottom of **Figure 10**) through panning, and different audio effects are applied to each of the channel strips, so that even if the same material occurs in multiple channels, the output of each will be different. This could easily be adapted to a multi-speaker system beyond the stereo field, in which case the

[1] Due to time restraints during composition and preparation, these two columns were not implemented.

grid controller could take on a much greater sense of spatial configuration and a different scoring paradigm would potentially be needed.

The process of the creation of all material from one source allows a great sense of sonic unity to be achieved, with a great deal of sonic variation and gestural intensity. As the score is navigated by the computer performer, a sense of unfolding and contracting is perceived, and indeed, by looking at the node connections (**Figure 7**) it is possible to see that there are points of convergence upon specific nodes, suggesting a possible expansion and contraction of the sound world that is produced by the computer performer.

4. CONCLUSIONS & FUTURE CONSIDERATIONS

Despite the limitations of direct interactivity discussed above, the composition *Dead Dirt* demonstrates a novel approach to mapping movement across an 8x8 grid controller to create a structured-improvisation drawing on various scoring paradigms and acoustic instruments with pre-composed electronics. As far as the author is aware at the time, due to the grid controller's primary association with more popular styles of music, a scoring paradigm for it has not been explored and yet it is an utterly unique method of controlling a computer system. A spatio-temporal approach to scoring seems the most useful based on the idiosyncratic design of the controller being a 2-dimensional space (X and Y) with the potential to conceive of a beginning at any point on the 64 pads without regard for hierarchical pitch structures. Put simply, treating the controller as an instrument in its own right and looking at its qualities and how best to notate for them.

Further experiments are needed to work towards greater interactivity between computer performer and their material, if that were desired by the composer, and also to look at how these novel scoring paradigms might translate into more popular styles of music, particularly electronic dance music, hip hop, and so on.

To conclude, the scoring system was highly successful for the context it was needed for, and has stimulated further considerations into how to improve on the paradigm, as well as locating many interesting convergences of what might be considered quite disparate practices both within computer music and outside of it.

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